

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

Dental Clinic Management and Recording System

In partial fulfillment for the requirements in Capstone 1 and Research Subject

John Marion Alcazar – BSIT 3
Ronald Sotomil Jr. – BSIT 3
Carl Emmanuel Ledesma – BSIT 3
Gloria Mar Anastacio – BSIT 3
Jhon Glenard Asencio – BSIT 3

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Dental clinics play a vital role in maintaining oral health by providing preventive, diagnostic, and treatment services to patients. These services include routine check-ups, dental cleaning, tooth restoration, and other specialized procedures that require accurate patient information and proper record management. Efficient handling of patient data is essential to ensure continuity of care and high-quality service delivery.

Despite advancements in technology, many small and medium-sized dental clinics still rely on traditional methods such as handwritten records, logbooks, and physical filing systems. These manual processes are often inefficient and prone to human error. As the number of patients increases, managing large volumes of paper-based records becomes more difficult, leading to disorganization and delays in retrieving information.

Based on interviews conducted with **Honeytooth Dental Clinic and Amparo Dental Clinic**, both clinics currently depend on manual systems for recording patient information and managing daily operations. Communication with patients is primarily done through Facebook Messenger and walk-in visits, while all records are documented manually and stored in physical files. This setup creates challenges in tracking patient history, managing appointments, and maintaining accurate records.

One of the most significant problems identified is the difficulty in retrieving old patient records. Records that date back several years are often hard to locate, which can delay treatment and affect the quality of care provided to patients. Additionally, manual systems increase the risk of data loss due to misplacement, damage, or deterioration of physical documents.

Furthermore, manual record-keeping consumes a considerable amount of time and effort for clinic staff. Tasks such as searching for patient files, updating records, and organizing documents can be time-consuming, reducing overall productivity. The lack

of a centralized system also makes it difficult to monitor clinic operations, generate reports, and ensure data accuracy.

With the rapid advancement of information technology, many healthcare institutions have adopted computerized systems to improve efficiency and accuracy. Digital systems allow for better data organization, faster retrieval of information, and enhanced security. These systems also support real-time updates and accessibility, enabling healthcare providers to deliver more effective services.

In response to these challenges, this study proposes the development of a Dental Clinic Management and Recording System. The system aims to digitize patient records, streamline clinic operations, and provide a centralized platform for managing information. By implementing this system, dental clinics can improve efficiency, reduce manual workload, and enhance the overall quality of patient care.

1.2 Statement of the Problem

Based on the interview conducted with the staff of **Honeytooth Dental Clinic** and **Amparo Dental Clinic**, the current process of managing patient records and clinic operations relies on manual documentation using handwritten records and paper files. This approach creates several challenges in organizing, retrieving, and monitoring important patient information.

This study seeks to address the following problems:

1. Patient records and clinic information are currently managed using manual and unorganized systems, making them difficult to maintain, update, and monitor efficiently.
2. The use of manual documentation increases the risk of errors, incomplete data entries, and potential loss of important patient records, which may affect the quality of service provided.
3. Daily clinic operations, such as recording patient information and tracking patient visits, are time-consuming and inefficient when handled through manual processes.
4. Retrieving and searching for old patient records requires considerable time and effort, especially when dealing with large volumes of physical files.
5. The absence of a centralized and computerized system makes it difficult for clinic staff to properly track, manage, and access patient data in an organized and efficient manner.

1.3 Objectives of the Study

1.3.1 General Objective

To design and develop a Dental Clinic Management and Recording System that provides an efficient, accurate, and centralized platform for managing patient records and clinic operations while minimizing manual workload and improving overall service delivery.

1.3.2 Specific Objectives

Specifically, the study aims to:

1. To analyze the current workflow of the dental clinics in managing patient records and daily operations.
2. To design a user-friendly system that will efficiently manage and organize patient records.
3. To develop a secure and centralized digital database for storing patient information.
4. To implement an efficient search and retrieval feature for faster access to patient records.
5. To minimize the time spent on manual processes in clinic operations.
6. To enhance the accuracy, integrity, and security of patient data within the system.

1.4 Scope and Delimitation

The study focuses on the development of a **Dental Clinic Management and Recording System** designed to improve the management of patient records and clinic operations. The system will include patient registration, record management, appointment tracking, search and retrieval of patient information, and an administrative dashboard for clinic staff.

The system will use a database to store patient information and clinic records in a centralized and organized manner. It will be developed using web-based technologies to allow easy access and management of data within the clinic.

The study will not cover online payment integration, advanced diagnostic tools, or integration with external healthcare systems. It will also not include large-scale deployment across multiple clinics. The application will be developed as a prototype for academic purposes only.

1.5 Significance of the Study

The proposed system will benefit the following:

- **Dental Clinics (Honeytooth Dental Clinic and Amparo Dental Clinic)** – The system will provide a more organized and efficient way of managing patient records, improving overall clinic operations.
- **Clinic Staff** – It will reduce manual workload and minimize errors in recording and retrieving patient information, leading to increased productivity.
- **Patients** – Patients will receive faster and more accurate services, as their records can be easily accessed and properly maintained.
- **Administrators** – The system will offer a centralized database that allows easier monitoring, reporting, and management of clinic data, supporting better decision-making.
- **Future Researchers and Developers** – This study will serve as a reference for future system development related to healthcare and digital record management.

1.6 Definition of Terms

- **Dental Clinic Management System** – A computerized system designed to manage patient records, appointments, and clinic operations efficiently.
- **Patient Record** – A collection of personal and medical information of a patient, including dental history, treatments, and visit details.
- **Manual System** – A traditional method of managing data using paper-based records and handwritten documentation.
- **Database** – An organized collection of data stored electronically, allowing easy access, retrieval, and management of information.
- **Data Accuracy** – The degree to which patient information is correct, complete, and free from errors within the system.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Foreign studies highlight the increasing use of digital systems in dental clinic operations. The development of a dental clinic management system was shown to improve patient record handling and appointment scheduling, resulting in reduced manual errors and increased efficiency (Zawawi & Ibrahim, 2023).

In addition, digital transformation in dental clinics has been found to enhance productivity and usability, especially in replacing paper-based records with computerized systems (Ho et al., 2024). These systems allow easier access to patient information and improve overall clinic workflow.

A web-based dental clinic management system was also developed to provide faster and more accurate healthcare services, particularly in managing patient data and clinic processes (Wardhana et al., 2024). The study emphasized that real-time data processing is essential in modern healthcare environments.

Furthermore, a smart dental care system known as DentaLink integrates appointment scheduling and centralized patient records, demonstrating that digital systems reduce delays and improve coordination compared to manual processes (Jadhav et al., 2025).

Electronic Health Record (EHR) systems have also been shown to improve the quality of care by allowing healthcare providers to access and update patient data efficiently (Sidek & Martins, 2017). This supports better decision-making and enhances patient service delivery.

2.2 Local Studies

Local and regional studies reveal that many dental clinics still rely on manual systems, which often result in inefficiencies and data management issues. A study on dental reservation management systems showed that implementing an information system improves accuracy and speeds up clinic operations (Nugraha & Kuswara, 2020).

In the Philippine context, a web-based dental clinic management system developed in Lucena City demonstrated improvements in handling patient records, appointment scheduling, and inventory management (Tagon & Largo, 2018).

The system helped reduce errors and improved service efficiency. Additionally, integrating advanced features such as chatbots into dental clinic systems has been found to enhance automation and accessibility of services (Alejandrino, 2023). This type of system enables better communication between patients and clinic staff.

These studies indicate that transitioning from manual to computerized systems is essential for improving clinic operations and patient satisfaction in local settings.

2.3 Related Literature

A dental clinic management system is a type of information system designed to manage administrative and clinical processes such as patient records, scheduling, and reporting. Digital systems improve accessibility and usability of patient information, allowing clinics to operate more efficiently (Ho et al., 2024).

Web-based systems enable real-time data processing and improve decision-making in healthcare services (Wardhana et al., 2024). These systems are particularly useful in managing large amounts of patient data and ensuring timely service delivery.

Process-based approaches in system development emphasize the importance of analyzing workflows to identify inefficiencies and improve clinic performance (Nery-da-Silva et al., 2019). This ensures that the system design aligns with actual clinic operations.

Moreover, modern clinic systems are developed using structured methodologies such as the Software Development Life Cycle (SDLC), which ensures proper planning, design, and implementation. Data integrity and security are also critical factors, as healthcare systems must protect sensitive patient information while maintaining accuracy and consistency.

2.4 Synthesis of the Reviewed Literature

The reviewed foreign and local studies consistently highlight the limitations of manual dental clinic systems, including inefficiency, data inaccuracies, and difficulty in accessing records. These issues negatively affect both clinic operations and patient satisfaction.

Foreign studies focus on advanced system features such as automation, integration, and real-time data processing, while local studies emphasize practical solutions for improving clinic efficiency. Despite these differences, both perspectives agree on the benefits of computerized systems.

The literature also stresses the importance of usability, accessibility, and data security in system development. These elements are essential in ensuring that the system is effective and user-friendly.

Overall, the reviewed studies support the development of a Dental Clinic Management and Recording System that integrates patient management, appointment scheduling, and record-keeping into a single platform. This system aims to address existing problems and improve the efficiency, accuracy, and quality of dental clinic services.

CHAPTER 3

3.1 Research Design

The study uses a developmental research design, which focuses on designing, developing, and evaluating a Dental Clinic Management and Recording System. This approach is appropriate as it aims to create a functional solution that improves the management of patient records and clinic operations.

3.2 System Development Methodology

The study will utilize the Agile Development Model, which is an iterative and flexible approach to system development. Agile focuses on continuous improvement, collaboration, and adaptability, making it suitable for developing systems that require frequent updates and user feedback.

consisting of the following phases:

1. Requirements Analysis -

This phase involves gathering system requirements through interviews, observation, and document analysis. The researchers identify the needs of the clinic, including patient record management, appointment scheduling, and reporting.

2. System Design -

In this phase, the system architecture, database structure, and user interface are designed. Diagrams such as Data Flow Diagrams (DFD) and Entity Relationship Diagrams (ERD) are created to visualize the system structure.

3. Development (Iteration Cycles) -

The system is developed in small increments or iterations. Each iteration includes coding, testing, and feedback to ensure that the system meets user requirements.

4. Testing -

The developed system undergoes various testing procedures, including functional testing, performance testing, and security testing to ensure reliability and accuracy.

5. Deployment (Prototype) -

The system is presented to users for evaluation. Feedback is collected and used to improve the system in future iterations.

6. Maintenance and Improvement -

Although the system is a prototype, improvements may be implemented based on user feedback to enhance performance and usability.

3.3 System Architecture

The system uses a client-server architecture where users access a web interface while the server processes and stores data in a centralized database, ensuring efficient, organized, and reliable record management.

3.4 Tools and Technologies Used

- Frontend: HTML, CSS, JavaScript
- Backend: PHP / Node.js
- Database: MySQL
- Development Tools: Visual Studio Code

3.5 Data Gathering Techniques

- Interviews and Questionnaires
- Document Review (patient records and logbooks)
- Observing Clinic Workflow

3.6 Testing Procedures

- Performance Testing
- Security Testing
- User Acceptance Testing

3.7 Evaluation Criteria

- Compatibility
- Maintainability
- Data Integrity
- Security

- Functionality
- Usability

Supporting Diagrams (Chapter 3)

Data Flow Diagram (DFD – Level 0)

Patient / Staff → Dental Clinic System → Database

- Staff registers patient or updates records
- System processes patient information, appointment, and treatment details
- System generates bills and processes payments
- Records and billing information are stored and retrieved from the database

Entity Relationship Diagram (ERD)

Entities:

- User (UserID, FullName, Username, Password, Role, Email, Phone, IsActive)
- Patient (PatientID, PatientCode, FirstName, LastName, MiddleName, DateOfBirth, Gender, Contact, Address)
- Service (ServiceID, ServiceName, Description, Duration, Price)
- Appointment (AppointmentID, PatientID, ServiceID, Date, Time, Status, HandledBy)
- DentalRecord (RecordID, PatientID, AppointmentID, ServiceID, Diagnosis, TreatmentDone, VisitDate)
- Bill (BillID, PatientID, AppointmentID, ServiceID, AmountDue, AmountPaid, PaymentMethod, Status)
- Schedule (ScheduleID, DayOfWeek, OpenTime, CloseTime, SlotDuration)
- BlockedDate (BlockedDateID, Date, Reason)
- Notification (NotificationID, UserID, Title, Message, Type, IsRead)
- AuditLog (LogID, UserID, Action, Module, RecordID, Date)

Relationships:

- Patient schedules Appointment
- Patient undergoes Treatment
- Staff manages Appointment
- Staff records Treatment
- Admin manages Staff Accounts
- Admin supervises Patient Records
- Appointment generates Bill
- Staff processes Payment

Use Case Diagram

Actors:

- Staff
- Administrator
- Patient

Use Cases:

- Register Patient
- Update Patient Record
- Schedule Appointment
- Record Treatment
- Generate Reports (Admin / Staff)
- Manage Staff Accounts (Admin)
- Create Bill / Invoice
- Process Payment (Cash, GCash, Bank)
- View Payment History)

End of Proposal